

D482

Secure Network Design

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A clinic hosts a public patient-portal web server and its internal records database on the same flat network. The portal was redesigned by an outside agency last year. What is the best design change?

- A. Put both behind one firewall on the same subnet.
- B. Have the agency redesign the portal again.
- C. Move the database into the DMZ next to the web server.
- D. Move the web server into a DMZ, keep the database on an internal segment, and allow only the web tier's database port through the firewall.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which pair of traffic flows does a well-designed DMZ usually permit?

- A. DMZ hosts to any internal host on any port.
- B. Internet to the DMZ web server on HTTPS, and the DMZ web server to one internal database port.
- C. No traffic at all may ever enter the DMZ.
- D. Internet directly to the internal database.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

At a logistics company, VPN users can reach every internal server once connected. The CIO notes the VPN uses strong AES-256 encryption. Which change moves toward zero trust?

- A. Check user identity and device health on each request and grant access only to the specific applications each role needs.
- B. Require longer VPN passwords.
- C. No change; the encryption is strong.
- D. Trust users on the office LAN automatically as well.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A new branch switch arrives with Telnet enabled, default admin credentials, SNMPv2c community “public,” and all ports up. It was shipped overnight from the vendor’s warehouse. Which set of steps best hardens it?

- A. Hold it in quarantine because overnight shipping may have damaged it.
- B. Keep Telnet but set a long password.
- C. Change only the admin password.
- D. Disable Telnet and use SSH, change default credentials, move to SNMPv3, and shut unused ports.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Why is SSH preferred over Telnet for managing network devices?

- A. Telnet is encrypted, but SSH also compresses.
- B. SSH is faster.
- C. SSH encrypts the session, including credentials.
- D. SSH needs no authentication.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A 300-employee company wants staff Wi-Fi with individual accountability plus a guest network. The office has glass walls and an open floor plan. Which design fits?

- A. An open guest network bridged to the staff VLAN.
- B. One WPA2-Personal passphrase for everyone.
- C. WPA3- or WPA2-Enterprise with 802.1X for staff; guests on a separate, isolated VLAN with internet-only access.
- D. Hide the SSID and use MAC filtering as the main control.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Every Monday morning a branch loses network access for about 20 minutes, and the team reboots the core switch to restore it. The branch manager is new. What should root cause analysis do?

- A. Replace the switch, since new hardware always fixes outages.
- B. Use logs and a timeline to find what happens each Monday, fix that cause, and confirm the outage stops.
- C. Schedule a reboot every Monday at 7 a.m.
- D. Train the new branch manager.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A draft firewall policy for a 50-person design studio lists specific block rules followed by “allow all” at the end. The studio moves large video files every day. How should it be redesigned?

- A. Deny all traffic with no exceptions.
- B. Allow only the traffic the studio needs and end with a default deny.
- C. Keep “allow all” and add more block rules above it.
- D. Put “allow all” at the top so video is not slowed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A firewall has (1) deny TCP from any to any on port 22, then (2) allow TCP from 10.0.0.5 to any on port 22, then an implicit deny. The admin at 10.0.0.5 cannot connect by SSH. What is the fix?

- A. Move rule 2 above rule 1.
- B. Change rule 1 to allow.
- C. Remove the implicit deny.
- D. Add a third rule that allows 10.0.0.5.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A company runs two independent internet links in active-active mode, each 99% available. One comes from a local fiber provider installed last spring. What is the combined availability?

- A. 100%, since redundancy removes all downtime.
- B. 98.01%
- C. 99.99%
- D. 99%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Traffic must pass through a firewall (99.9% available) and then a router (99.5% available) in series. What is the path's availability, to one decimal place?

- A. 99.5%
- B. About 99.9995%
- C. 99.7%
- D. About 99.4%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A scan finds three issues: (A) CVSS 9.8 on an internet-facing VPN appliance; (B) CVSS 9.8 on a lab server with no network path to production or the internet; (C) CVSS 5.3 on an internal printer. The scan took six hours. Which should be fixed first?

- A. A
- B. B
- C. C
- D. A and B equally, since equal scores always mean equal priority.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A clinic hosts a public patient-portal web server and its internal records database on the same flat network. ~~The portal was redesigned by an outside agency last year.~~ What is the best design change?

Target word: segment the tiers **Decides:** Public and sensitive systems share one network.

A	Dropped element	Leaves them able to talk freely to each other.
B	Camouflage	The page design is not the network risk.
C	Inverted operation	Exposes the most sensitive system.
D	Correct	Limits what an attacker can reach if the public server is compromised.

2. Answer B Concept 1 • transfer

Which pair of traffic flows does a well-designed DMZ usually permit?

Target word: reverse: design to allowed flows **Decides:** Only specific, needed flows cross zones.

A	Dropped element	Drops the restriction that makes a DMZ useful.
B	Correct	Each hop is narrow and needed.
C	Absolute language	Then the public service could not work.
D	Inverted operation	Skips the DMZ entirely.

3. Answer A Concept 2

At a logistics company, VPN users can reach every internal server once connected. ~~The CIO notes the VPN uses strong AES-256 encryption.~~ Which change moves toward zero trust?

Target word: zero-trust change **Decides:** Access is granted by network location.

A	Correct	Zero trust verifies continuously and limits reach.
B	Adjacent category	Strengthens login but keeps the flat access.
C	Camouflage	Encryption protects the tunnel, not what users can reach.
D	Inverted operation	Adds more location-based trust.

4. Answer D Concept 3

A new branch switch arrives with Telnet enabled, default admin credentials, SNMPv2c community “public,” and all ports up. ~~It was shipped overnight from the vendor’s warehouse.~~ Which set of steps best hardens it?

Target word: hardening steps **Decides:** Several insecure defaults are present.

A	Camouflage	Shipping speed is not a security issue.
B	Confusion set	Telnet sends the password in cleartext.
C	Dropped element	Leaves cleartext management and open ports.
D	Correct	Addresses every insecure default.

5. Answer C Concept 3 • transfer

Why is SSH preferred over Telnet for managing network devices?

Target word: reason behind the control **Decides:** Encryption of the session.

A	Inverted operation	Telnet is not encrypted.
B	Adjacent category	Speed is not the reason.
C	Correct	Telnet sends everything in cleartext.
D	Confusion set	SSH still requires authentication.

6. Answer C Concept 4

A 300-employee company wants staff Wi-Fi with individual accountability plus a guest network. ~~The office has glass walls and an open floor plan.~~ Which design fits?

Target word: wireless design **Decides:** Per-user authentication and guest separation.

A	Inverted operation	Gives guests a path to internal systems.
B	Dropped element	A shared key gives no individual accountability.
C	Correct	Each user authenticates individually; guests cannot reach internal systems.
D	Confusion set	Both are easy to get around.

7. Answer B Concept 5

Every Monday morning a branch loses network access for about 20 minutes, and the team reboots the core switch to restore it. ~~The branch manager is new.~~ What should root cause analysis do?

Target word: RCA approach **Decides:** Recurring issue cleared by a workaround.

A	Absolute language	Without evidence the cause may be elsewhere.
B	Correct	RCA removes the cause instead of repeating the workaround.
C	Adjacent category	Automates the workaround and hides the cause.
D	Camouflage	The manager's tenure is unrelated.

8. Answer B Concept 6

A draft firewall policy for a 50-person design studio lists specific block rules followed by "allow all" at the end. ~~The studio moves large video files every day.~~ How should it be redesigned?

Target word: rule-base design **Decides:** The policy defaults to allow.

A	Absolute language	Blocks the work the studio needs.
B	Correct	Anything not explicitly needed is blocked.
C	Adjacent category	Still permits anything not listed.
D	Camouflage	Chasing the video detail turns off every other rule.

9. Answer A Concept 6 · transfer

A firewall has (1) deny TCP from any to any on port 22, then (2) allow TCP from 10.0.0.5 to any on port 22, then an implicit deny. The admin at 10.0.0.5 cannot connect by SSH. What is the fix?

Target word: fix rule order **Decides:** First match wins; rule 1 matches first.

A	Correct	The specific allow must come before the broad deny.
B	Inverted operation	Opens SSH to everyone.
C	Dropped element	Rule 1 still blocks the admin, and all other traffic opens.
D	Adjacent category	It would still sit below the deny.

10. Answer C Concept 7

A company runs two independent internet links in active-active mode, each 99% available. ~~One comes from a local fiber provider installed last spring.~~ What is the combined availability?

Target word: parallel availability **Decides:** Both must fail at once: 0.01×0.01 .

A	Absolute language	Both links can still fail together.
B	Inverted operation	0.99×0.99 is the series formula.
C	Correct	$1 - (0.01 \times 0.01) = 0.9999$.
D	Dropped element	Ignores the second link.

11. Answer D Concept 7 · transfer

Traffic must pass through a firewall (99.9% available) and then a router (99.5% available) in series. What is the path's availability, to one decimal place?

Target word: series availability **Decides:** Both must be up: multiply.

A	Dropped element	Takes only the weaker device.
B	Inverted operation	Uses the parallel formula.
C	Confusion set	Averages the two values.
D	Correct	$0.999 \times 0.995 = 0.994005$.

12. Answer A Concept 8

A scan finds three issues: (A) CVSS 9.8 on an internet-facing VPN appliance; (B) CVSS 9.8 on a lab server with no network path to production or the internet; (C) CVSS 5.3 on an internal printer. ~~The scan took six hours.~~ Which should be fixed first?

Target word: priority **Decides:** Same severity, but A is exposed to the internet.

A	Correct	Critical severity plus internet exposure is the highest risk.
B	Dropped element	Same score, but isolation lowers its risk.
C	Inverted operation	Lowest severity and internal only.
D	Absolute language	Exposure changes risk even when scores match.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Segmentation and the DMZ	Put public-facing services in a DMZ and internal data in separate segments; allow only the specific flows each tier needs.	One flat network behind one firewall, or moving sensitive data into the DMZ.	A public server and an internal database on the same network.
2	Zero trust	Never grant trust because of network location; verify identity, device, and context for each request and grant least-privilege access per application.	Treating a connected VPN or office LAN as trusted.	“Once connected, users can reach everything.”
3	Device hardening	Replace defaults, disable unused services and ports, use encrypted management (SSH, SNMPv3, HTTPS), and restrict management access.	Changing only the password.	A new device with Telnet, default credentials, or SNMP “public.”
4	Wireless security design	Use WPA3 or WPA2-Enterprise with 802.1X for staff so each user is authenticated; isolate guests on their own VLAN with internet-only access.	A shared passphrase, hidden SSIDs, or MAC filtering as the main control.	“Individual accountability” and “guest access.”
5	Root cause analysis	Find and fix the underlying cause (logs, timelines, five whys) rather than repeating a workaround.	Scheduling the workaround or replacing hardware without evidence.	A recurring outage cleared by a reboot.
6	Firewall rule design	Rules are read top down, first match wins; end with default deny and allow only what is needed.	Allow-all at the bottom, or a broad rule above a specific one.	An ordered rule list that does not behave as intended.
7	Redundancy and availability	In parallel, $A = 1 - (1 - a_1)(1 - a_2)$; in series, $A = a_1 \times a_2$.	Multiplying for parallel parts, or claiming 100%.	Two links or devices with percent availability.
8	Prioritizing vulnerabilities	Rank by risk: severity combined with exposure and asset value, not by score alone.	Treating equal CVSS scores as equal priority.	Several findings with scores and different exposure.

Course scope (public description): Secure Network Design provides the foundational knowledge and skills to design secure physical and logical network architectures for wired and wireless networks. Course topics include the characteristics of a secure network, techniques to securely configure network devices, network segmentation strategies, root cause analysis, and mitigation approaches based on industry best practices. The course also offers hands-on experience in network vulnerability analysis and network configuration.